

DNREC Permitting Intelligence

Helping DNREC reviewers deliver faster, clearer, regulation-backed feedback to vendors.

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Department of Natural Resources and Environmental Control

The problem with permitting

Every septic permit DNREC approves is a check against **groundwater contamination** when implementing a new septic system.

- Manual review takes **2-3 hours per permit**
- Every hour spent on desk review is an hour employees **aren't out doing in-person inspections**
- This can result in **stalled construction and repair**. Stalled repairs means a **failing tank can keep leaking sewage**.

WHYY

WATERSHED | WHYY NEWS CLIMATE DESK

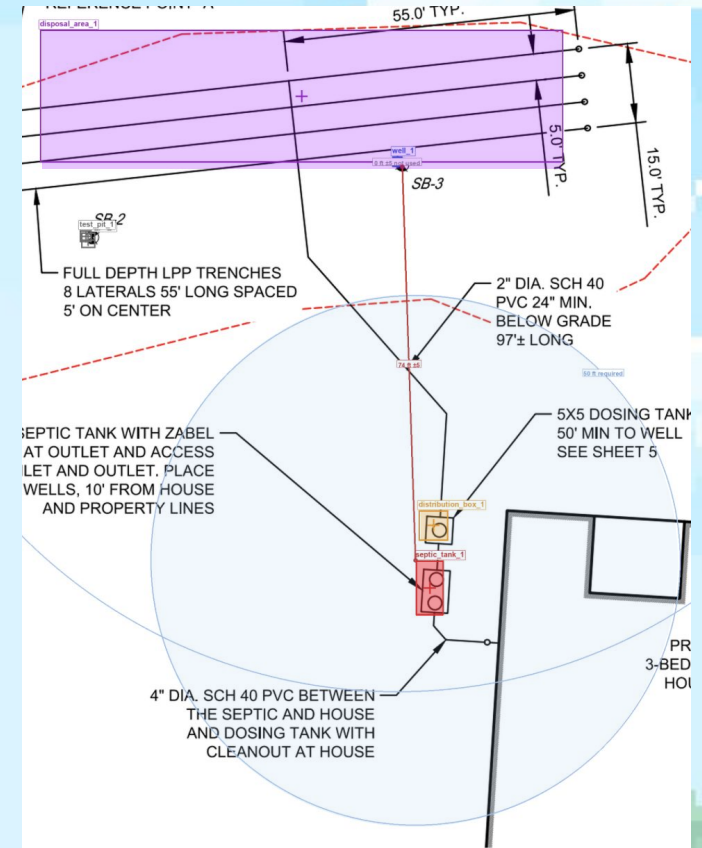
Human waste has been detected in Southern Delaware waterways. Home septic systems may be to blame

Homeowners are not required to inspect their septic systems until they sell their home, meaning deteriorating systems could go unnoticed.

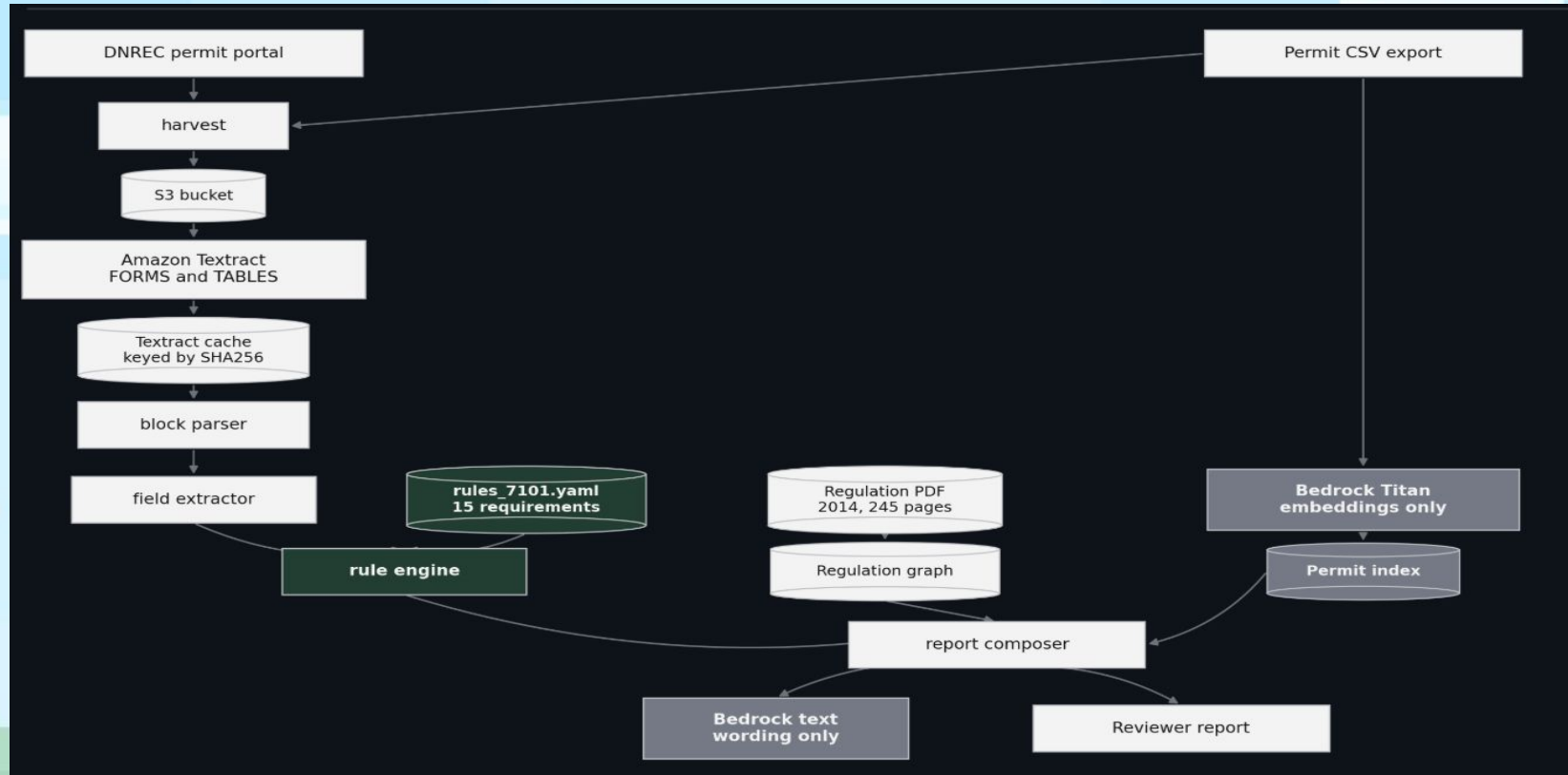
Our Solution

An employee can drop a permit, and be able to understand the review at a fraction of the time.

- Every finding traced to the **section, page, and quoted sentence of the 2014 regulation it came from.**
- If a deficiency is found, the system creates clear, **reviewer-approved** feedback showing the vendor what needs correction.
- AI identifies the site plan and measures **well-to-septic** distance using the drawing scale.



The Pipeline



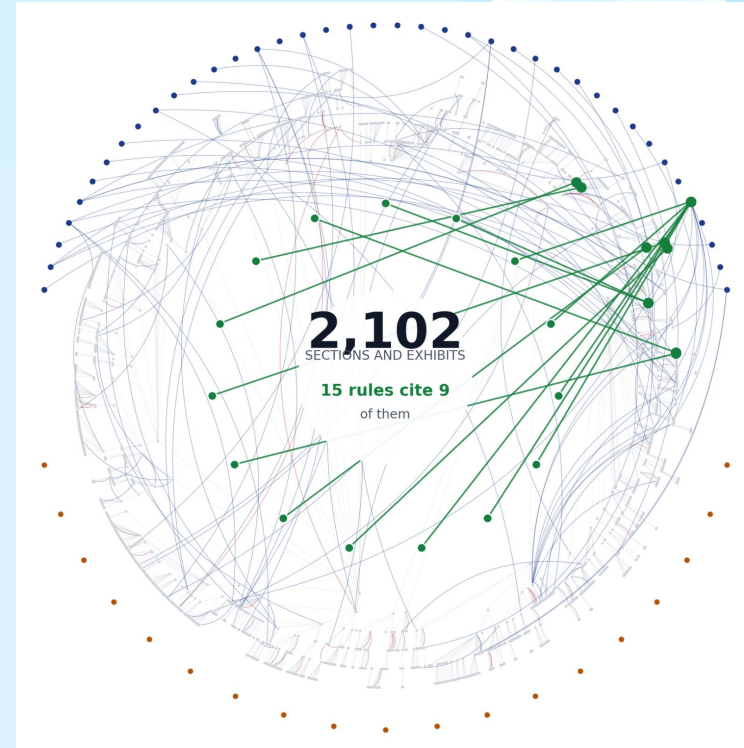
Responsible AI & Data

- Used public permit packets, **Delaware regulations**, and **non-sensitive geographic data**
- Owner information is **excluded from the AI assistant**.
- **Deterministic rules** —not AI— produce review findings.
- Every finding **includes a regulation citation**.
- Reviewers verify findings and approve corrections before vendors receive them.

Beyond an API call: We built document extraction, verified rules, a regulation knowledge graph, grounded AI safeguards, automated testing, and reviewer-controlled annotated PDFs.

Compute Platform & Learning

- **Amazon Textract:** Extracted text, forms, tables, and checkboxes from permit PDFs.
- **Amazon Bedrock:** Interpreted site-plan images and extracted details that standard OCR could not reliably understand.
- **Amazon Neptune:** Connected regulations, definitions, exhibits, and citations.
- **Google Vertex AI — Gemini 2.5 Flash:** Powered the grounded reviewer assistant.
- **Amazon S3:** Supported permit-document processing.



From Prototype & Production

Works today

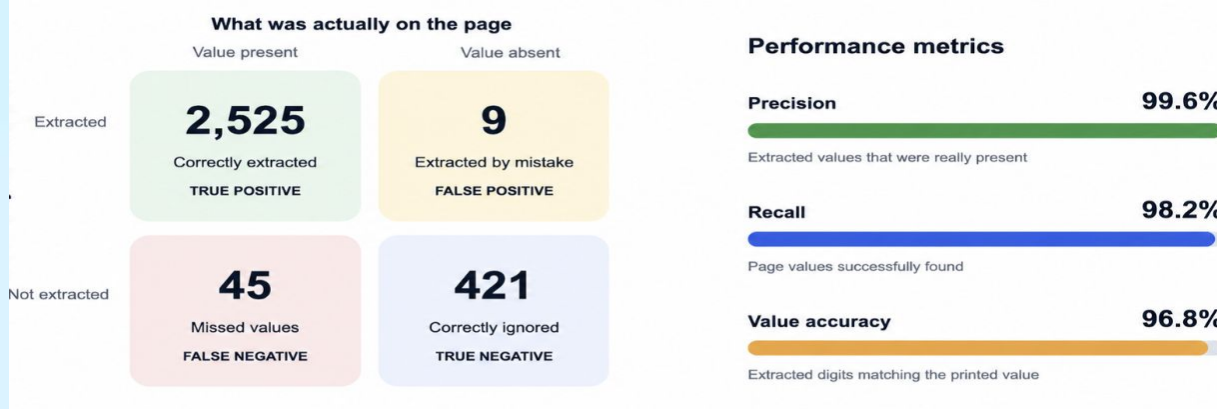
- Reads permits and site plans
- Applies 15 verified checks
- Provides cited findings
- Creates vendor-ready corrections

Limitations

- Limited rules and test permits
- Some measurements require human review
- Not yet production-scale

Next steps

- Expand and validate rules
- Test more permit types
- Add agency security and monitoring
- Pilot with DNREC reviewers



Impact for Delaware

- The **manpower** that was previously spent reviewing permits can now be spent **on-site** to review **septic system installations**.
- More consistent **review quality** and **consistent environmental safeguards**
- The application can be **scaled and personalized** for different permits [wells, wastewater collection, etc]

We see this system growing for Delaware.

Key Takeaways

- Complex permit packets **create slow reviews and unclear corrections.**
- AI-assisted document analysis, deterministic rules, and cited findings **support, not replace, the reviewer**
- Reviewers can quickly see what passed, what is missing, and what vendors need to correct.
- This **results in faster reviews**, clearer vendor communication, and fewer resubmission cycles.